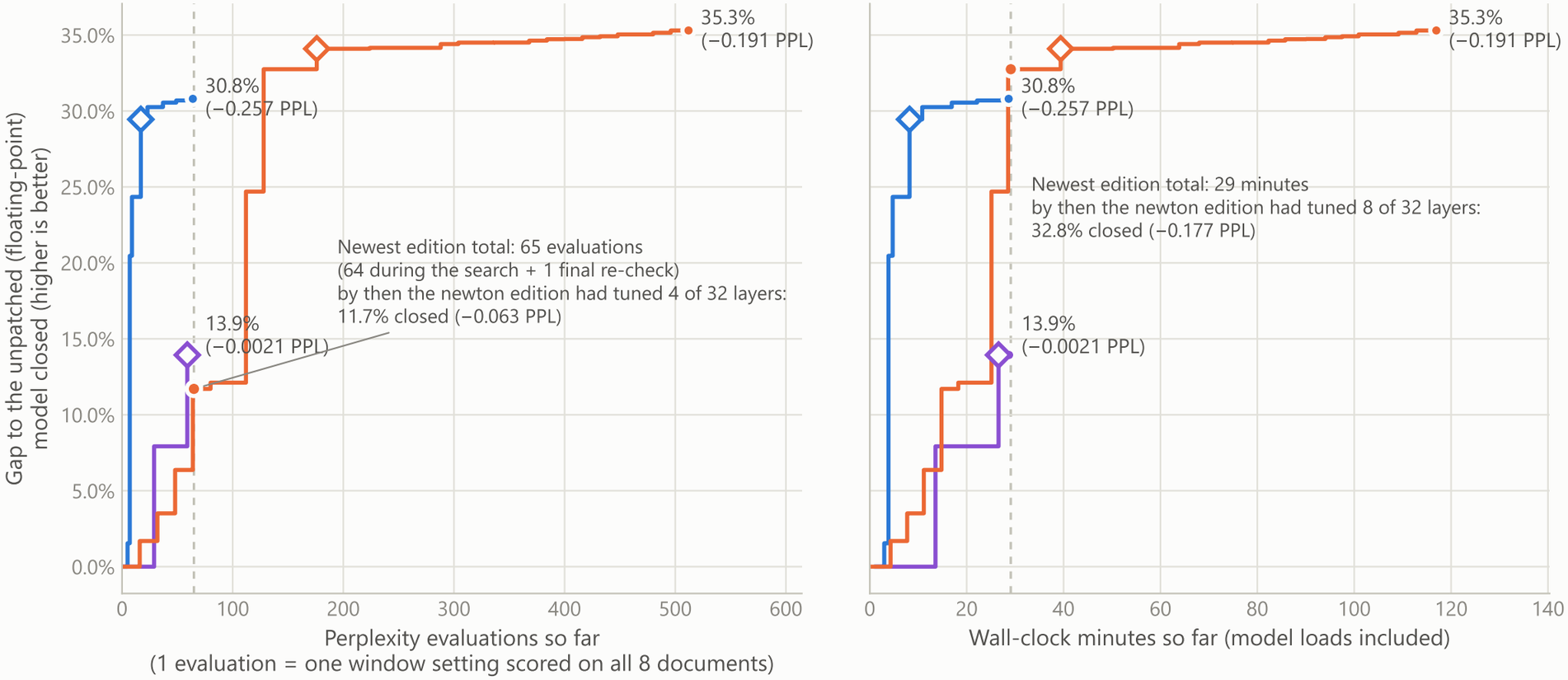
Search progress against cost

Question: does automating the per-layer search recover a similar share of accuracy for less compute?
Shows: one full pass costs the Newest edition 65 evaluations and 29 min, the newton edition 512 and 117 min (~8× and ~4×). Accuracy is only comparable within the same kernel.
Each line is one pass over all 32 layers. Each step up is a layer whose window the search changed. Up and to the left is better.
Gap closed = (start PPL − PPL now) / (start PPL − unpatched floating-point PPL). 0% = no better than the uniform starting window, 100% = as good as the unpatched model.



Hollow diamonds: where each run first reached 95% of its own final gain. Newest edition, unfixed kernel: evaluation 17 of 64 (layer 8); Newest edition, fixed kernel: evaluation 59 of 64 (layer 29); newton edition: evaluation 176 of 512 (layer 11).
That point is only known after the fact: the search cannot tell in advance that later layers will not help, which is why it runs every layer.

- Newest edition, unfixed kernel (mask leak), 2×H200, time budget ended during layer 32
- Newest edition, fixed kernel, 1×H200, time budget ended during layer 32
  2 of 63 candidate windows accepted, each ~0.001 PPL; with 4 batches about 6 would pass by chance (estimate), so this is not evidence of a real gain
- newton edition (lab's original manual per-layer tuning, Fig. 7 method), unfixed kernel, 2×H200